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A Short Introduction to Causal Inference (Part 1)

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Outline

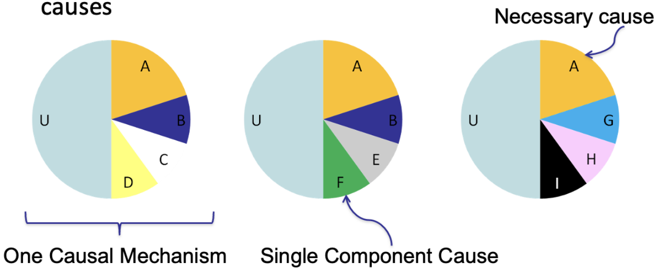
Thanks to Dr. Michael Hudgens at the University of North Carolina Chapel Hill for materials used to develop these lectures! Some examples drawn from What If Book

Theories of Causation

- Divine retribution: Imbalance in body humors caused by air, water, land, stars; spontaneous generation
- Miasma: Disease transmitted by miasmas or clouds clinging to earth's surface
- Germ Theory and Henle-Koch Postulates
- Web of causation

Modern Theories of Causation

- Causal pies (sufficient, necessary, component causes)
- Hill's causal guidelines
- Structural causal models
- Neyman Rubin Potentil Outcomes Framework (focus for today!) (Neyman, 1923; Rubin 1970s)
  - 3 sufficient causal complexes, each having 5 component causes



1.1 Potential Outcomes

- Dichotomous (binary) treatment A taking values 0 (untreated), 1 (treated)
- Dichotomous outcome Y taking values 0, 1
- Potential outcome $Y^{a=1}$ outcome that would have been observed if, possibly counter to fact, treatment $a = 1$ was received
- Potential outcome $Y^{a=0}$ defined similarly
  aka counterfactuals
- E.g., A is receive heart transplant or not, Y is die or not 5 days later
- Neyman (1923), popularized by Rubin (1970s); sometimes referred to as the "Neyman-Rubin causal model"

1.1 Causal Consistency

- Observed outcome Y related to potential outcomes by *causal consistency* theorem (Pearl *Epidemiology* 2010)

$$Y = Y^{a=1}A + Y^{a=0}(1 - A)$$

- Not to be confused with the statistical concept of consistency (i.e., convergence in probability)

1.1 Example

Table 1.1 of Potential Outcomes and Table 1.2 of Observed Data

Patient	$Y^{a=0}$	$Y^{a=1}$
Rheia	0	1
Kronos	1	0
Demeter	0	0
Hades	0	0
Hestia	0	0
Poseidon	1	0
Hera	0	0
Zeus	0	1
Artemis	1	1
Apollo	1	0
Leto	0	1
Ares	1	1
Athena	1	1
Hephaestus	0	1
Aphrodite	0	1
Cyclope	0	1
Persephone	1	1
Hermes	1	0
Hebe	1	0
Dionysus	1	0

Patient	A	Y
Rheia	0	0
Kronos	0	1
Demeter	0	0
Hades	0	0
Hestia	1	0
Poseidon	1	0
Hera	1	0
Zeus	1	1
Artemis	0	1
Apollo	0	1
Leto	0	0
Ares	1	1
Athena	1	1
Hephaestus	1	1
Aphrodite	1	1
Cyclope	1	1
Persephone	1	1
Hermes	1	0
Hebe	1	0
Dionysus	1	0

SUTVA (Fine Points 1.1-1.2)

- No interference is part of SUTVA (Rubin 1980)
- Stable Unit Treatment Value Assumption
    1. No interference
    2. Only one version of treatment and one version of no treatment (control)

Or if there are multiple versions of treatment, they are irrelevant (Cole and Frangakis 2008, VanderWeele 2009, Pearl 2010)
See Fine Point 1.2 regarding multiple versions of treatment

1.2 Average Causal Effects

- There is an *average causal effect* in the (super-)population if

$$\Pr[Y^{a=1} = 1] \neq \Pr[Y^{a=0} = 1]$$

or more generally (i.e., for a non-dichotomous Y)

$$E[Y^{a=1}] \neq E[Y^{a=0}]$$

- There is no average causal effect in the population if

$$\Pr[Y^{a=1} = 1] = \Pr[Y^{a=0} = 1]$$

This is implied by, but does not imply, the sharp null

- E.g., treatment may have an effect on some individuals, but no average effect

1.3 Measures of Causal Effects

- Causal risk difference (RD)

$$\Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1] =$$
$$E[Y^{a=1}] - E[Y^{a=0}] \; (= 0 \text{ under null})$$

  The righthand side can be written as $E[Y^{a=1} - Y^{a=0}]$ and is often referred to as the average causal effect

- Causal risk ratio (RR)

$$\frac{\Pr[Y^{a=1} = 1]}{\Pr[Y^{a=0} = 1]} \; (= 1 \text{ under null})$$

- Causal odds ratio (OR)

$$\frac{\Pr[Y^{a=1} = 1] / \Pr[Y^{a=1} = 0]}{\Pr[Y^{a=0} = 1] / \Pr[Y^{a=0} = 0]} \; (= 1 \text{ under null})$$

- Because these measures describe the causal effect of treatment, referred to as *effect measures*

1.5 Measures of Association

- Associational risk difference (RD)

$$\Pr[Y = 1|A = 1] - \Pr[Y = 1|A = 0] =$$
$$E(Y = 1|A = 1) - E(Y = 1|A = 0)$$

- Associational risk ratio (RR)

$$\frac{\Pr[Y = 1|A = 1]}{\Pr[Y = 1|A = 0]}$$

- Associational odds ratio (OR)

$$\frac{\Pr[Y = 1|A = 1]/\Pr[Y = 0|A = 1]}{\Pr[Y = 1|A = 0]/\Pr[Y = 0|A = 0]}$$

- If $A \perp\!\!\!\perp Y$ (independent) then assoc. RD equals 0, assoc. RR and OR equal 1

Example

- Consider again Tables 1.1 and 1.2
- No average causal effect (ignoring sampling variability) because
$$\widehat{\Pr}[Y^{a=1} = 1] = \frac{10}{20} = \widehat{\Pr}[Y^{a=0} = 1]$$
- However, there is an association
$$\widehat{\Pr}[Y = 1|A = 1] = \frac{7}{13} \neq \frac{3}{7} = \widehat{\Pr}[Y = 1|A = 0]$$
- Note, HR write Pr instead of $\widehat{\Pr}$ for simplicity.
- Association is not causation

Randomized Experiments

- Consider a randomized experiment where individuals are assigned treatment $a = 1$ or $a = 0$ randomly, i.e., independent of their potential outcomes
- Full exchangeability $\{Y^{a=1}, Y^{a=0}\} \perp\!\!\!\perp A$
- Exchangeability $Y^a \perp\!\!\!\perp A$ for $a = 0, 1$
- Full exchangeability implies exchangeability

Exchangeability

- Under mean exchangeability

$$E[Y^{a=1}] - E[Y^{a=0}] = E[Y|A=1] - E[Y|A=0]$$

  Similarly, the causal RR and OR equal the associational RR and OR
- Thus in randomized experiments, association is causation!
- Note that exchangeability for $a = 0, 1$,

$$Y^a \perp\!\!\!\perp A$$

  is different from $Y \perp\!\!\!\perp A$.

In a randomized experiment, (full) exchangeability will hold, where as $Y \perp\!\!\!\perp A$ will not hold if A has a causal effect on Y

Conditional Randomization

Z is some baseline covariate

Patient	Z	A	Y
Rheia	0	0	0
Kronos	0	0	1
Demeter	0	0	0
Hades	0	0	0
Hestia	0	1	0
Poseidon	0	1	0
Hera	0	1	0
Zeus	0	1	1
Artemis	1	0	1
Apollo	1	0	1
Leto	1	0	0
Ares	1	1	1
Athena	1	1	1
Hephaestus	1	1	1
Aphrodite	1	1	1
Cyclope	1	1	1
Persephone	1	1	1
Hermes	1	1	0
Hebe	1	1	0
Dionysus	1	1	0

Conditional Randomization

- Conditionally randomized experiments can be viewed as two separate marginal experiments
- Thus, conditional exchangeability holds

$$Y^a \perp\!\!\!\perp A|Z = z \text{ for } a, z = 0, 1$$

where is written as $Y^a \perp\!\!\!\perp A|Z = z$ for $a = 0, 1$ for notational ease

- In the language of missing data, potential outcomes are MCAR in a marginally randomized experiment and MAR in a conditionally randomized experiment

Randomized Experiment Paradigm

An observational study can be conceptualized as a conditionally randomized experiment under the following three conditions:

1. Values of treatment under comparison correspond to well-defined interventions (Sections 3.4-3.5)
2. Conditional probability of receiving every value of treatment, though not decided by investigators, depends only on the measured covariates (Section 3.2)
3. Conditional probability of receiving every value of treatment is greater than zero, i.e., positive (Section 3.3)

Exchangeability

- In observational studies, when treatment is not randomly assigned by the investigators, it may be dubious to assume exhangeability $Y^a \perp\!\!\!\perp A$
- We might, however, be willing to assume conditional on some set of covariates Z that conditional exchangeability holds $Y^a \perp\!\!\!\perp A|Z$
- For example, suppose donor hearts are scarce, so that doctors tend to direct heart transplants to patients who need them most

  Thus, we may not be willing to assume $Y^0 \perp\!\!\!\perp A$

  However, we may be willing to assume $Y^0 \perp\!\!\!\perp A|Z$ where Z is the indicator whether patient is in critical condition

Conditional Exchangeability (CE)

- Unfortunately, it is not possible to verify $Y^a \perp\!\!\!\perp A|Z$ because, in order to do so, one would need to test

$$\Pr[Y^a = 1|A = a, Z = z] = \Pr[Y^a = 1|A \neq a, Z = z]$$

  But this is not possible because Y^a is never observed for individuals with $A \neq a$ (i.e., right side is not identifiable)
- Thus causal inference in observational studies often relies on expert knowledge to select Z in order to ensure CE plausible
- CE will not hold if there exist unmeasured independent predictors U of outcome such that probability of receiving treatment A depends on U within strata of Z. For this reason, $Y^a \perp\!\!\!\perp A|Z$ often referred to as the *no unmeasured confounders* assumption.

Conditional Exchangeability

If it is not possible to verify $Y^a \perp\!\!\!\perp A | Z$, what can we do?

1. Rely on expert knowledge to assess plausibility of CE
2. Posit the existence of U and conduct sensitivity analysis
3. Derive bounds

Positivity

- Positivity is defined to be

$$\Pr[A = a | Z = z] > 0 \text{ for all } z \text{ such that } \Pr[Z = z] > 0$$

- In Table 3.1 positivity holds because there are individuals at both levels of treatment ($A = 0$ and $A = 1$) for each level of the covariate Z (0 and 1)
- Positivity would not hold if an individual with critical condition at baseline $Z = 1$ always gets treatment, i.e., $\Pr[A = 0 | Z = 1] = 0$
- Unlike conditional exchangeability, positivity can sometimes be empirically verified

Well-defined interventions

- Up until now we have assumed there are only two versions of treatment, $a = 1$ and $a = 0$, and hence two potential outcomes, $Y^{a=0}$ and $Y^{a=1}$, per individual (recall this was part of SUTVA)
- However it may be there are different versions of treatment $a = 1$
- E.g., "heart transplant" might entail different surgeons, different pre-op procedures, etc
- These different versions of treatment could result in different potential outcomes

Well-defined interventions

- Causal consistency and no multiple versions of treatment often conflated
- Consistency is an axiom/theorem, not an assumption (Pearl 2010)
- No multiple versions of treatment is an assumption
- This assumption can be relaxed by more finely defining treatment levels (and thereby increasing the number of potential outcomes considered, see VanderWeele 2009)

Causation and Prediction

- What if we are not willing to assume positivity, conditional exchangeability, and well-defined interventions?
- There is always prediction (association)
  Obesity (A) may predict (or be associated with) mortality risk (Y)
- This does not imply causation, just as carrying lighter in pocket (A) being predictive of lung cancer (Y) does not imply carrying a lighter causes cancer
- However, associations may be helpful in generating hypothesis. Why is obesity associated with mortality risk? Is it diet? Exercise? Other well-defined interventions?
- From a public health or scientific viewpoint we may want to go beyond associations to attempt to understand why such associations exist

The structure of confounding

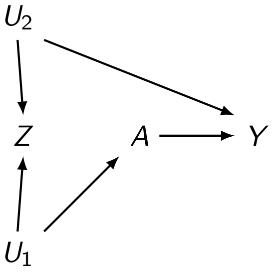
- Genetic factors: Suppose DNA sequence A affects risk of developing trait Y
- Suppose there is another DNA sequence Z that also affects Y
- Z and A are associated due to a common cause U, such as ethnicity
- Figure 7-3

$$
\begin{tikzcd}
Z \arrow[rr, bend left] & A \arrow[r] & Y \\
U \arrow[u] \arrow[ur] & &
\end{tikzcd}
$$

Again there is confounding because U is a common cause of A and Y

Confounders

- The traditional definition of confounder (statistical rather than structural)
  1. Associated with treatment
  2. Associated with outcome conditional on treatment
  3. Not on the causal pathway between treatment and outcome
- Consider Figure 7-4



- Traditional definition suggests Z is a confounder (and adjust for Z)

Confounding and exchangeability

- How do DAGs relate to potential outcomes/counterfactuals?
- How does the structural definition of confounding relate to conditional exchangeability?
- Exchangeability $Y^a \perp\!\!\!\perp A$ equivalent to no confounding
- Conditional exchangeability $Y^a \perp\!\!\!\perp A|Z$ equivalent to Z being a sufficient set for confounding adjustment (i.e., no unblocked backdoor paths)
- Single World Intervention Graphs (SWIGs) make this connection between potential outcomes and graphs explicit (Richardson and Robins 2013)

The structure of selection bias

- An association created as a result of the process by which individual are selected into the analysis is referred to as selection bias
- Selection bias encompasses various biases
- One example is given in Figure 8-1, where A is treatment, Y is outcome and C is a common effect of A and Y

$$\begin{tikzcd} A \arrow[r] \arrow[rr, bend left] & Y \arrow[r] & \boxed{C} \end{tikzcd}$$

- Example: Suppose A is folic acid (0/1) given to a woman shortly after conception, Y is fetus cardiac malformation (0/1) during the first two months of pregnancy, and C is death (0/1) prior to birth (miscarriage)
- Suppose the study conditioned on survival after birth $C = 0$

The structure of selection bias

- Consider Figure 8-2

$$\begin{tikzcd} A \arrow[r] \arrow[rr, bend left] & Y \arrow[r] & C \arrow[r] & \boxed{S} \end{tikzcd}$$

- Suppose S denotes parental grief (0/1) and only non-grieving parents $S = 0$ are willing to participate in the study
- Again there are two sources of association between A and Y
  1. The open path $A \rightarrow Y$
  2. The open path $A \rightarrow C \leftarrow Y$, which is open because we have conditioned on the descendent S of the collider C on the path between A and Y
- In the first two examples, we conditioned on a common effect of A and Y; however, selection bias can arise in other settings

Example of selection bias

- Differential loss-to-follow-up, informative censoring (in both randomized experiments and observational studies)
- Missing data bias, nonresponse bias (of complete case analysis)
- Healthy worker bias
- Self-selection bias, volunteer bias
- Selection affected by treatment received after study entry (e.g., lifetime exposure as inclusion criteria)
- Case-control studies

9.1 Measurement error

- Key idea: There is *measurement bias* when the association between treatment and outcome is weakened or strengthened as a result of the process by which the study data are measured
- Until now, we have assumed all variables (treatment, outcome, covariates, etc.) measured without error
- This assumption is often not realistic
- Example: Treatment A will often be measured with error: physician incorrectly records prescribed drug, patient does not comply with treatment
- Denote measured treatment A^* and true (possibly unobserved) treatment A

Measurement error

- In Figure 9.1

$$\begin{tikzcd}
U_A \arrow[d] & \\
A^* & \\
A \arrow[u] \arrow[r] & Y
\end{tikzcd}$$

- True treatment A affects outcome Y and measurement treatment A^*
- U_A includes all factors other than A that determine A^*
- We refer to U_A as *measurement error*
- Note that U_A does not need to included on the causal DAG
- Goal: Draw inference about effect of A on Y despite not observing A or U_A

Standardization

THINK BIG WE DO℠

- Consider design 2, a conditionally randomized experiment
- Then

$$
\begin{aligned}
E[Y^a] &= \sum_z E[Y^a | Z = z] \Pr[Z = z] \\
&= \sum_z E[Y^a | A = a, Z = z] \Pr[Z = z]
\end{aligned}
$$

  where the second equality holds by conditional exchangeability
- Therefore, by causal consistency

$$
E[Y^a] = \sum_z E[Y | A = a, Z = z] \Pr[Z = z]
$$

  Note the right side of this equation, aka standardized mean, is identifiable from the observable random variables (Z, A, Y)

Standardization

- Under conditional exchangeability, the standardized mean equals the counterfactual mean had all individuals in population received treatment a
- Thus causal effects can be identified under conditional exchangeability, e.g., the casual RR equals

$$\frac{\Pr[Y^{a=1}=1]}{\Pr[Y^{a=0}=1]} = \frac{\sum_z E[Y=1|A=1,Z=z]\Pr[Z=z]}{\sum_z E[Y=1|A=0,Z=z]\Pr[Z=z]}$$

- In the usual scenario where we are not ignoring sampling variability, this suggests a consistent estimator based on plugging-in observed proportions

$$\frac{\sum_z \widehat{\Pr}[Y=1|A=1,Z=z]\widehat{\Pr}[Z=z]}{\sum_z \widehat{\Pr}[Y=1|A=0,Z=z]\widehat{\Pr}[Z=z]}$$

- Similarly for the causal RD and OR

Inverse Probability Weighting

- Consider the following Horwitz-Thompson-type estimator of $E[Y^a]$

$$\frac{1}{n}\sum_{i=1}^{n}\frac{I[A_i = a]Y_i}{\Pr[A_i = a|Z_i]}$$

- This estimator is unbiased when $\Pr[A = a|Z = z]$ is known for all a, z (as in a conditional randomized experiment) and conditional exchangeability holds
- $\Pr[A = 1|Z = z]$ is the *propensity score*
- Notes: (i) notation $\Pr[A = a|Z]$ is different from notation of HR (Technical Point 2.2); (ii) $\Pr[A = a|Z]$ is a random variable whereas $\Pr[A = a|Z = z]$ is a constant

Doubly robust estimator

1. Fit treatment model $\pi(L, \gamma)$ for $\Pr[A = 1|Z = z]$ (e.g., using logistic regression) and obtain estimated propensity scores

$$\pi(Z, \hat{\gamma}) = \widehat{\Pr}[A = 1|Z_i]$$

2. Fit outcome model $m_a(Z, \beta_a)$ for $E[Y|A = a, Z]$ and compute predicted potential outcome for individual

$$m_a(Z, \beta_a) = \hat{E}[Y|A = a, Z_i]$$

3. Construct DR estimators (Lunceford and Davidian *Stat in Med*, 2004)

$$\tilde{E}[Y^1] = n^{-1} \sum_i \frac{A_i Y_i - \{A_i - \pi(Z_i, \hat{\gamma}\} m_1(Z_i, \hat{\beta}_1)}{\pi(Z_i, \hat{\gamma})}$$

$$\tilde{E}[Y^0] = n^{-1} \sum_i \frac{(1 - A_i) Y_i + \{A_i - \pi(Z_i, \hat{\gamma}\} m_0(Z_i, \hat{\beta}_1)}{1 - \pi(Z_i, \hat{\gamma})}$$

Doubly robust estimator

- $\tilde{E}[Y^a]$ is CAN of $E[Y^1]$ provided either the treatment model or the outcome model is correctly specified, but not necessarily both!
- If $\hat{\gamma} \xrightarrow{p} \gamma^*$ and $\hat{\beta}_1 \xrightarrow{p} \beta_1^*$, then

$$
\begin{aligned}
E[Y^1] \xrightarrow{p} & E\left(\frac{AY^1 - \{A - \pi(Z, \gamma^*)\}m_1(Z_i, \beta_1^*)}{\pi(Z_i, \gamma^*)}\right) \\
&= E\left(Y^1 - \frac{\{A - \pi(Z, \gamma^*)\}m_1(Z_i, \beta_1^*)}{\pi(Z_i, \gamma^*)}\right) \\
&= E[Y^1] - E\left(\frac{\{A - \pi(Z, \gamma^*)\}m_1(Z_i, \beta_1^*)}{\pi(Z_i, \gamma^*)}\right)
\end{aligned}
$$

- Assume $Y^1 \perp\!\!\!\perp A|Z$, the last expectation equals zero if $\pi(Z_i, \gamma^*) = \Pr[A = 1|Z]$ (i.e., propensity score model is correct) or if $m_1(Z_i, \beta_1^*) = E[Y|A = 1, Z]$ (i.e., outcome model is correct)

Weighted Regression

- Assume OR model $g\{[Y|L, A=1]\} = m_1(Z;\alpha_1)$, but use weighted regression with inverse probability weights $e^{-1}(Z;\hat{\beta})$
- Write $s_1(Z_i;\hat{\alpha}_1) = g^{-1}\{m_1(Z;\hat{\alpha}_1)\}$, obtain an OR estimator

$$\hat{\mu}_1 = \frac{1}{N}\sum_{i=1}^{N} s_1(Z_i;\hat{\alpha}_1) = \frac{1}{N}\sum_{i=1}^{N} s_1(Z_i;\hat{\alpha}_1) + \frac{1}{N}\sum_{i=1}^{N} \frac{A_i}{e(Z_i;\hat{\beta})}\{Y_i - s_1(Z_i;\hat{\alpha}_1)\}$$

- Again $\hat{\alpha}_1$ solves the score equation

$$0 = \sum_{i=1}^{N} \frac{A_i}{e(Z_i;\hat{\beta})}\{Y_i - s_1(Z_i;\hat{\alpha}_1)\}$$

- Weighted regression estimator $\hat{\mu}_1$ is also DR (Robins et al, 2007)

Doubly robust estimator

- Simulation study
- $n = 500$, four scenarios, 200 data sets per scenario
- $Z \sim N(0,1)$; $A \sim Bern(logit^{-1}(Z))$; $Y^1 \sim N(Z,1)$; $Y^0 \sim N(Z,1)$
- Empirical bias

Scenario	IPW	ParaG	DR
Both correct	0.00	0.00	0.00
Only propensity score correct	0.00	0.82	0.01
Only outcome correct	0.82	0.00	0.00
Neither correct	0.82	0.82	0.82

For incorrect models, Z^2 used instead of Z.

DR: Cautionary Remarks

- Empirical evidence suggests outcome model plays a more prominent role than the PS model:
  - Misspecification of the outcome model leads to much larger bias than misspecification of the PS model
  - Even if the PS model is correct, a misspecified outcome model usually lead to notable bias
- Empirically, the extra protection from the propensity score augmentation is somewhat limited
- Kang and Schafer (2007, Stat Sci) gives an example where moderate misspecification of both PS and outcome model (very likely scenario in practice) leads to large bias and variance of DR

Transmission Reduction Intervention Project

- Sociometric network-based study of injection drug users and their contacts in Athens, Greece from 2013 to 2015.
- Intervention: **Community alerts**
  Outcome: **HIV risk behavior at 6-month follow up**

Baseline interview
- 277 participants
- 29 alerted

→ six-month time period →

Follow-up interview
- 57 missing
- 92 risk behavior

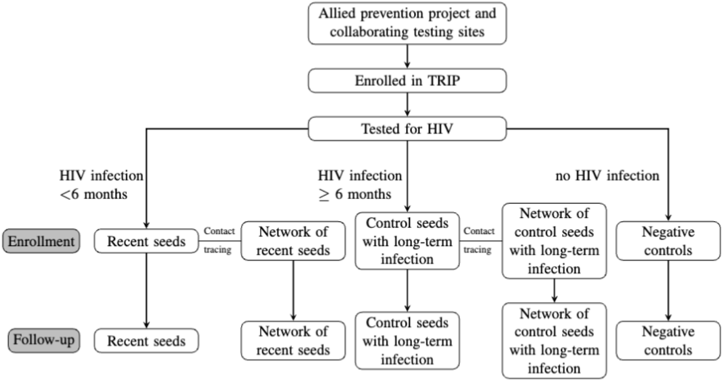


Figure 1: Recruitment in TRIP (Pampaka et al. 2021)

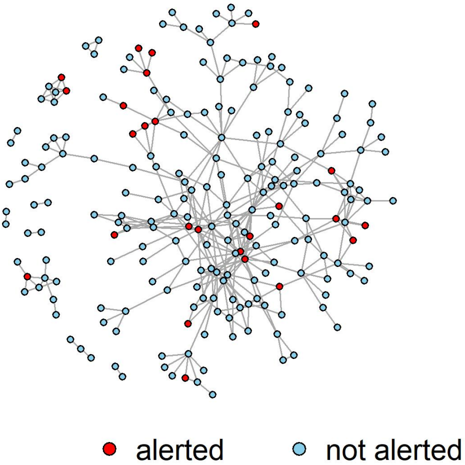


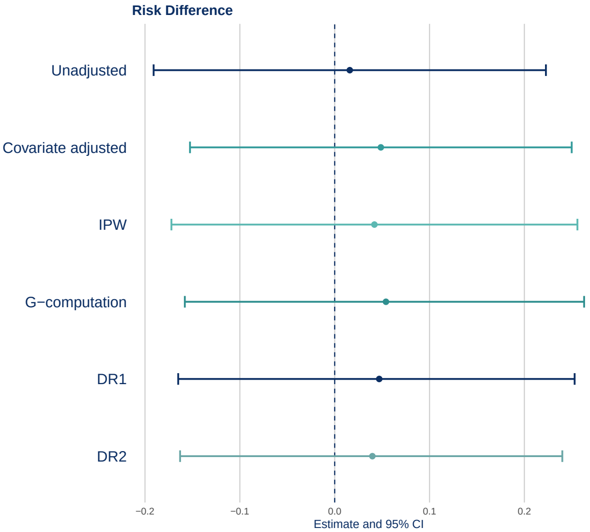
Figure 2: TRIP network with isolates removed. There are 216 participants and 362 links in the network.

Analysis under SUTVA in TRIP

- Assuming SUTVA and independence of observations, we assess the causal effect of community alerts on HIV risk behavior.
- We will estimate risk ratios using a log-Binomial regression (unadjusted), then adjust for measured baseline confounders using IPW, g-computation and doubly-robust estimators.
- Why are the assumptions of SUTVA and IID dubious in this study?

Network Characteristics	Nodes	216
	Edges	362
	Components	10
	Average Degree (SD)	3.35 (2.75)
	Density	0.016
	Transitivity	0.25
	Assortativity	0.25
Baseline Visit		
Community alert	Exposed	25 (13%)
	Not Exposed	191 (88.4%)
HIV Status	Positive	113 (53%)
	Negative	103 (48%)
Date of first interview	Before ARISTOTLE ended	110 (51%)
	After ARISTOTLE ended	106 (49%)
Education	Primary School or less	64 (30%)
	High School (first 3 years)	68 (32%)
	High School (last 3 years)	52 (24%)
	Post High School	32 (15%)
Employment status	Employed	33 (15%)
	Unemployed; looking for work	54 (25%)
	Can't work; health reason	101 (47%)
	Other	28 (13%)
Shared injection equipment in last 6 months	Yes	159 (74%)
	No	57 (26%)
Six-month Visit		
Outcome: sharing injection equipment at the 6-month visit	Yes	92 (43%)
	No	124 (57%)

RD Effect Estimates in TRIP (SUTVA)



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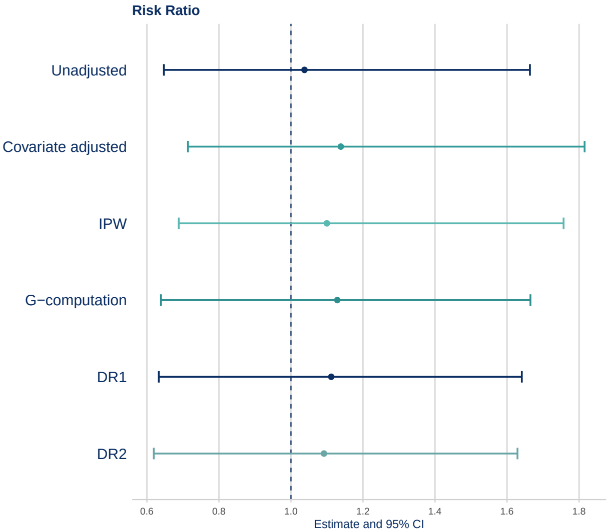
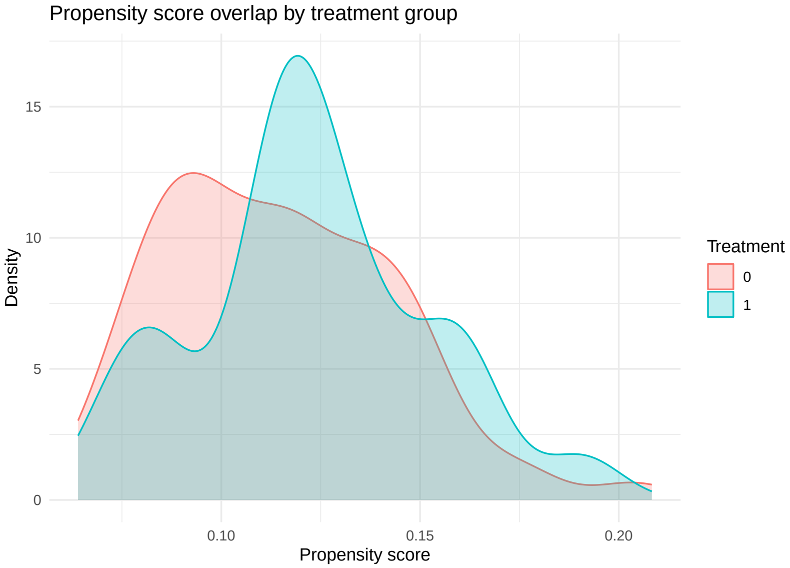
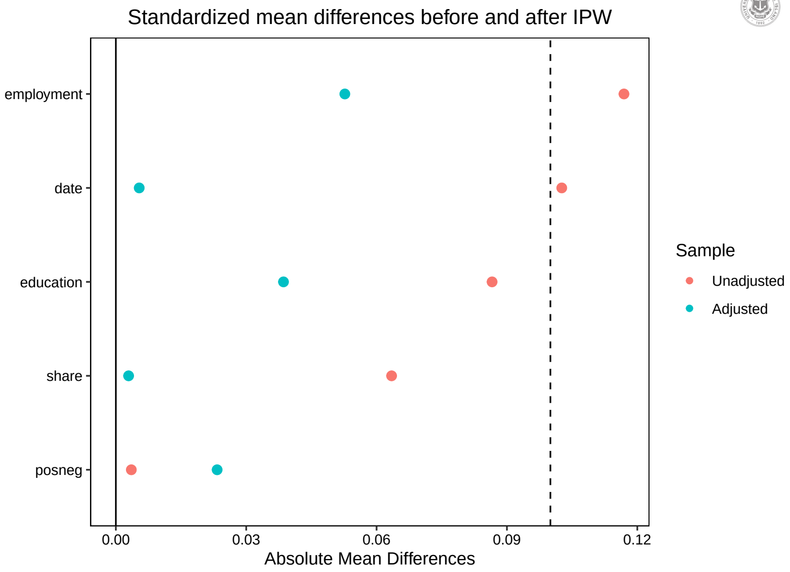



Propensity score overlap by treatment group
15
10
5
0
Density
0.10
0.15
0.20
Propensity score
Treatment
0
1


Standardized mean differences before and after IPW
employment
date
education
share
posneg
0.00
0.03
0.06
0.09
0.12
Absolute Mean Differences
Sample
Unadjusted
Adjusted

Textbooks

- Imbens, Guido W., and Donald B. Rubin. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press, 2015.
- Morgan, Stephen L., and Christopher Winship. *Counterfactuals and Causal Inference*. Cambridge University Press, 2014.
- Pearl, Judea. *Causality*. Cambridge University Press, 2009.
- Rosenbaum, Paul R. *Observational Studies*. Springer New York, 2002.

Suggested references for further reading

Buchanan, Ashley L., et al. "Worth the Weight: Using Inverse Probability Weighted Cox Models in AIDS Research." *AIDS Research and Human Retroviruses* 30.12 (2014): 1170-1177.

Cole, Stephen R., and Miguel A. Hernan. "Constructing inverse probability weights for marginal structural models." *American Journal of Epidemiology* 168.6 (2008): 656-664.

Hernan, Miguel A. "Does water kill? A call for less casual causal inferences." *Annals of Epidemiology* 26.10 (2016): 674-680.

Hudgens, Michael G., and M. Elizabeth Halloran. "Toward causal inference with interference." *Journal of the American Statistical Association* (2012), 832-842.

Little Roderick J., and Donald B. Rubin."Causal effects in clinical and epidemiological studies via potential outcomes: concepts and analytical approaches." *Annual Review of Public Health* 21.1 (2000): 121-145.

Lunceford, Jared K., and Marie Davidian. "Stratification and weighting via the propensity score in estimation of causal treatment effects: a comparative study." *Statistics in Medicine* 23.19 (2004): 2937-2960.

Robins, James M., Miguel Angel Hernan, and Babette Brumback. "Marginal structural models and causal inference in epidemiology." *Epidemiology* (2000): 550-560.

Web Resources

- Networks and Causal Inference for Public Health Research at URI `https://web.uri.edu/ncipher/`
- Hernan and Robins book `http://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/`
- Pearl's Causality Blog `http://causality.cs.ucla.edu/blog/`
- Gelman "Statistical Modeling, Causal Inference, and Social Science" `http://andrewgelman.com/`
- Society for Causal Inference `https://sci-info.org/`

Simulated Toy Data based on TRIP Data - 1

- Step 1: **Generate a network**
  - Calculating approximate total number of edges based on number of nodes and mean degree in true TRIP network, then use function `sample_gnm()` in R package `igraph` to generate a network with specifying number of nodes and number of edges.
- Step 2: **Simulate covariates**.
  - Binary: Simulate using binomial distribution with probability equals to the mean of the true covariate.
  - Education: Primary or less vs. high school or more; Employment: Not able able to work vs. other.
  - Continuous: Simulate using normal distribution with mean and standard error obtained from true covariate.

Simulated Toy Data based on TRIP Data - 2

- Step 3: **Simulate all potential outcomes**.
  - Fit outcome model based on true TRIP data: $Y_i \sim A_i + \frac{s_i}{d_i} + \boldsymbol{Z}_i$, with $\boldsymbol{Z}_i =$ c(posneg.x, date, share.x, edubin, employbin, age) is a 6×1 covariate vector, get regression coefficients.
  - Simulate all potential outcomes using coefficients obtained above using the formula:
    $Y_i \sim \text{Bern}(p = \text{logit}^{-1}(\beta_0 + \beta_1 A_i + \beta_2 \frac{s_i}{d_i} + \beta_3 A_i \frac{s_i}{d_i} + \boldsymbol{Z}_i^\top \boldsymbol{\beta}_Z))$,
    Simulate Y_i for all possible A_i and s_i.
- Step 4: **Simulate treatment**.
  - Fit treatment model based on true TRIP data, get coefficients, use these coefficients to simulate treatment.
    $A_i \sim \text{Bern}(p = \text{logit}^{-1}\left(\gamma_0 + \boldsymbol{Z}_i^\top \boldsymbol{\gamma}_Z\right))$
- Step 5: **Simulate "observed" outcome**.
  - Use A value generated in Step 4 and corresponding s, d count, use the following formula to simulate an "observed" outcome.

$$Y_i \sim \text{Bern}(p = \text{logit}^{-1}(\beta_0 + \beta_1 A_i + \beta_2 \frac{s_i}{d_i} + \beta_3 A_i \frac{s_i}{d_i} + \boldsymbol{Z}_i^\top \boldsymbol{\beta}_Z))$$

Publicly Available TRIP Data

Transmission Reduction Intervention Project, Athens site data: Nikolopoulos, Georgios K., Friedman, Samuel R., and Buchanan, Ashley L. Transmission Reduction Intervention Project (TRIP), Athens, Greece Site, 2013-2015. Inter-university Consortium for Political and Social Research [distributor], 2024-12-09. TRIP Data

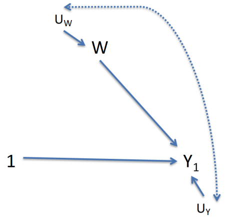
Causal Roadmap

- Framework for estimating causal parameters from real data
  1. State the research question and hypothetical experiment
  2. Define the causal model and parameter of interest
  3. Link the causal model to the observed data and define the statistical model
  4. Assess identifiability: link the causal effect to a parameter estimable from the observed data
  5. Choose and apply the estimator
  6. Derive an estimate of the sampling distribution (statistical uncertainty)
  7. Make inferences (interpret findings)

Causal Model: SCM 1

Generate counterfactuals by intervening on the causal model, set $A = 1$:

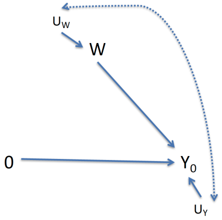
- $Z = f_W(U_Z)$
- $A = 1$
- $Y^1 = f_Y(Z, 1, U_Y)$
- Y^1: outcome if possibly-contrary to fact, the unit was exposed $(A = 1)$



Causal Model: SCM 2

Generate counterfactuals by intervening on the causal model, set $A = 0$:

- $Z = f_W(U_Z)$
- $A = 0$
- $Y^0 = f_Y(Z, 0, U_Y)$
- Y^0: outcome if possibly-contrary to fact, the unit was exposed $(A = 0)$



Thank you! Questions?

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